Dialectal Bias in Hate Speech Detection Systems

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Abstract. We audit two openly available classifiers, a hate speech detector and a toxicity detector, on three Twitter corpora: Davidson et al., the Twitter part of HateXplain and the US random sample of HateDay. Tweets are grouped by how closely their wording matches African-American English (AAE) with the dialect model of Blodgett et al. On tweets that humans labelled *neither hate nor offensive*, the AAE-aligned group is flagged more often by both classifiers in all three corpora (false positive rate gaps from +0.02 to +0.30, all intervals above zero), and a random AAE-aligned tweet gets a higher score than a random mainstream one with probability 0.62 to 0.75.

1 Introduction

Social media platforms rely on automatic classifiers to flag hateful content. This systems are usually trained and tested on text close to mainstream written English. Friedman and Nissenbaum call a system biased when it works well for the population it was built and tested on, but fails for users who differ from it. African-American English (AAE) is a rule-governed variety of English with its own grammar and vocabulary. Earlier work found that hate speech datasets, and the models trained on them, treat tweets associated with AAE differently. In this project we audit two openly available classifiers, on three different datasets.

1.1 Research Questions

This project will address the following research questions:

- Do the classifiers flag AAE-aligned tweets that are *not* hate more often than mainstream tweets?
- Does the gap appear for both classifiers, not only one?
- Does the gap appear in all three corpora?
- Does the gap stay when we use a different threshold?
- Does the gap stay when we define the groups in another way?
- Is the gap caused by the label mix of the two groups?
- Is the gap caused by profanity?
- Is the gap caused by tweet length?
- Do the words in the tweet explain the gap?
- Which words go with the false positives?

The hypothesis is that the classifiers flag more of the AAE-aligned tweets that are not hate than of the mainstream ones.

2 Background and Related Work

2.1 Automated Hate Speech Detection and the Problem of Offensive Language

Davidson et al. [2] collected tweets that match a hate speech lexicon and had crowd workers label them as *hate speech*, *offensive language* or *neither*. Their main point is that systems tend to confuse offensive language with hate speech. This distinction is central here. A classifier may disagree with the labels because its definition of hate differs, not because it is wrong. The main idea is that offensive language is not hate speech.

2.2 Demographic Dialectal Variation in Social Media: A Case Study of AAE

Blodgett et al. [1] built a mixed-membership model from geolocated tweets and Census demographics. For a tweet, it gives proportions over four demographically aligned language groups, one of which is African-American-aligned. The output describes how much a text resembles the usage of a group. It does not say who wrote it. In this project it is used as a measurement instrument to split tweets into AAE-aligned and mainstream groups, and this limit is stated wherever it matters.

2.3 The Risk of Racial Bias in Hate Speech Detection

Sap et al. [4] showed that AAE markers correlate with toxicity labels in hate speech datasets, that classifiers trained on these data propagate the correlation, and that annotators primed with dialect information label fewer tweets as offensive. Their results concern models *trained* on annotated corpora. In this project, it is used ready-made pretrained classifiers and it is evaluted on independent corpora. Following the usual reading of fairness as comparable error rates across groups, it was used the false positive rate (FPR) as the main quantity.

2.4 HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection

HateXplain [3] is a benchmark of about 20,000 posts from Twitter and Gab. It has majority labels (hateful, offensive, normal), the targeted community and the words that justify each label. This project uses its Twitter part as a second corpus, because the dialect model was built on Twitter. It differs from Davidson in collection, annotation and text normalisation, so a gap found in both is less likely to be a property of one dataset.

2.5 HateDay: Insights from a Global Hate Speech Dataset Representative of a Day on Twitter

HateDay [5] is a random sample of all tweets posted on a single day (21 September 2022), annotated for hate speech. Unlike Davidson, which was selected with a lexicon, it reflects what a platform actually sees. Most tweets are harmless and hate is rare. This project only uses the US part as a third corpus to test whether the gap seen in lexicon-selected data also appears in everyday tweets. The dataset is gated, so no tweet is printed, quoted or committed in this project.

3 Materials and Methods

3.1 Overview

The pipeline has three components with different roles, shown in Figure 1. The Davidson corpus provides the ground truth (human labels). The dialect predictor assigns each tweet to a group. The RoBERTa hate classifier labels each tweet as hate or not hate. The two models run independently on the tweet text and their outputs are joined at the evaluation step.

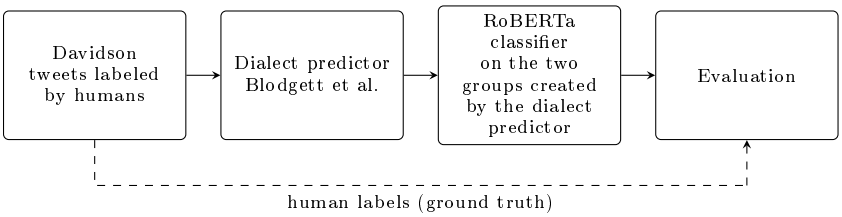


Fig. 1. The pipeline.

3.2 Davidson et al. dataset

In this project it was used the corpus of Davidson et al. [2]. This dataset has 24,783 tweets with the full text, each labelled by crowd workers as **hate speech** (1,430 tweets), **offensive language** (19,190) or **neither** (4,163). Since the classification is binary, it will only be mapped the *hate speech* class to the positive class (hate) and treat both *offensive language* and *neither* as not hate. This choice defines what counts as an error throughout the report, and in the Section 5 this will be further explained.

3.3 Blodgett et al.

To predict the dialect from the tweet, it was used the released model of Blodgett et al. [1]. This model it was ported from Python 2 to Python 3 without changing

the inference logic. The model has a vocabulary of 191,874 words and a count table that gives the usage of each word in the four language groups. The groups are *African-American*, *Hispanic*, *Asian* and *White*.

For a tweet, it tokenises the text with a Twitter-specific tokenizer, lowercases it, and keeps the in-vocabulary tokens. It then infers a probability distribution after seing the full tweet text. The result is a vector with four probability distributions, and we use the first one, the African-American distribution, as the *AAE score* of the tweet. Tweets with no in-vocabulary token, or with fewer than 20% in-vocabulary tokens, get no score. The score is a bag-of-words, t says how much of the wording resembles African-American-aligned usage, and it is not the probability that the author is African American.

We define two groups with thresholds on the AAE score. Tweets with a score of at least 0.6 are *AAE-aligned*, tweets with a score of at most 0.4 are *mainstream*, and the tweets in between are *ambiguous* and excluded. We also exclude 7 tweets that received no score. This leaves 17,900 tweets, of which 4,689 are AAE-aligned and 13,211 are mainstream.

3.4 Hate classifier (RoBERTa)

The audited classifier is. According to its model card, it is a RoBERTa-base model pretrained on about 58 million tweets and fine-tuned for binary hate speech detection on the TweetEval benchmark, that is, on HatEval. We run it as is, with no further training, through the Hugging Face `transformers` library.

Before inference, user mentions are replaced by `@user` and links by `http`, as in the preprocessing of the model, and inputs are truncated to 128 tokens. The predicted label is the one the model returns, which corresponds to the default decision threshold. The model was trained on HatEval, a corpus collected independently of the Davidson tweets.

3.5 Evaluation

All metrics treat hate as the positive class, and are computed separately for each group from the counts of true and false positives and negatives (TP, FP, FN, TN). Table 1 defines them. In this project the difference between the AAE-aligned and the mainstream value of a metric its called *gap*.

To test whether a gap was larger than sampling noise, it was used a non-parametric bootstrap. In each of 2,000 rounds, it was drawn each group again with replacement, with the same size as the original, it was then computed the metric on both resampled groups and record the gap. The 95% confidence interval was given by the 2.5th and 97.5th percentiles of the 2,000 gaps. The approximate two-sided p-value is twice the smaller of the shares of resampled gaps that are at least zero and below zero. The bootstrap measures only sampling noise, it does not account for errors of the dialect predictor or of the human labels. It was tested the gaps in F1 and in FPR.

Metric	Formula	Meaning
Precision	TP / (TP + FP)	Of the tweets flagged as hate, how many are hate?
Recall	TP / (TP + FN)	Of the hate tweets, how many were caught?
F1	2TP / (2TP + FP + FN)	Balance between precision and recall.
Accuracy	(TP + TN) / all	How many tweets are classified correctly?
FPR	FP / (FP + TN)	Of the tweets that are not hate, how many are flagged?

Table 1. Evaluation metrics.

3.6 Implementation

TODO TODO TODO

groups, with a precision of about 0.03 and an F1 of about 0.06. F1 is practically identical in the two groups. The other metrics are not. The recall is 0.588 for AAE-aligned tweets and 0.149 for mainstream tweets, and the FPR is 0.751 against 0.338. Accuracy is 0.262 against 0.623. The classifier is therefore much more willing to flag AAE-aligned tweets, both the ones that are hate and the ones that are not.

Metric	AAE-aligned	Mainstream	Gap
Precision	0.031	0.035	−0.004
Recall	0.588	0.149	+0.439
F1	0.058	0.056	+0.002
Accuracy	0.262	0.623	−0.361
False positive rate	0.751	0.338	+0.412

Table 3. Metrics at the default threshold. Gap = AAE-aligned minus mainstream.

4.3 Significance of the gaps

Table 4 shows the bootstrap results. The interval of the F1 gap contains zero and the gap is about as large as the noise (gap-to-noise ratio about 0.1), so there is no evidence of a difference in F1. The interval of the FPR gap is far from zero, and the gap is about 27 times the noise: the sign of the FPR gap did not change in any of the 2,000 resamples. The FPR of AAE-aligned tweets is 2.2 times that of mainstream tweets.

Gap in	Observed	95% bootstrap CI	p
F1	+0.002	$[-0.012, +0.015]$	0.73
FPR	+0.412	$[+0.397, +0.428]$	< 0.001

Table 4. Bootstrap test of the gaps (2,000 resamples, seed 42).

4.4 Reference points

Metrics are easier to judge against simple references (Table 5). The classifier flags 74.5% of all AAE-aligned tweets and 32.4% of all mainstream tweets, while the shares of hate speech according to the labels are 3.9% and 7.5%. The precision divided by the share of hate (the *lift*) is 0.79 and 0.46. A classifier that flags tweets at random would have a lift of 1, so at the default threshold the flagged tweets contain a smaller share of hate speech than the whole group does. A model that flags nothing would have an accuracy of 96.1% and 92.5%, against the 26.2% and 62.3% observed.

	AAE-aligned	**Mainstream**
Share of hate speech (labels)	3.9%	7.5%
Share of tweets flagged by the model	74.5%	32.4%
Precision lift (precision / share of hate)	0.79	0.46
Accuracy of a model that never flags	96.1%	92.5%
Hate tweets missed (1 − recall)	41.2%	85.1%

Table 5. Reference points for the metrics.

4.5 Sensitivity to the dialect threshold

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4.6 Which words?

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5 Discussion and Conclusion

5.1 What the results show?

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5.2 What explains the gap?

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5.3 Limitations

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5.4 Future work

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5.5 Conclusion

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